```
CREATE TABLE client_master (
    client_no VARCHAR2(6) PRIMARY KEY CHECK (client_no LIKE 'C%'),
    name VARCHAR2(30) NOT NULL,
    address1 VARCHAR2(30),
    address2 VARCHAR2(30),
    city VARCHAR2(15),
    state VARCHAR2(15),
    pincode NUMBER(6),
    balance_due NUMBER(10,2)
);
INSERT INTO client_master VALUES('C001','IvanBayross','P-
76','Worli','Bombay','Maharastra',400054,15000);
INSERT INTO client_master VALUES('C002','VandanaSatiwal','128','Adams
Street','Madras','TamilNadu',780001,0);
INSERT INTO client_master
VALUES('C003','PramadaJaguste','157','Gopalpur','Kolkata','West
Bengal',700058,5000);
INSERT INTO client_master
VALUES('C004','BasuNavindgi','A/12','Nariman','Bombay','Maharastra',400056,0)
;
INSERT INTO client_master VALUES('C005','Ravi
Sreedharan','B/34','Rajnagar','Delhi','Delhi',100001,2000);
INSERT INTO client_master VALUES('C006','Rukmini','Q-
12','Bandra','Bombay','Maharastra',400050,0);

CREATE TABLE product_master (
    product_no VARCHAR2(6) PRIMARY KEY CHECK (product_no LIKE 'P%'),
    description VARCHAR2(40) NOT NULL,
    profit_percent NUMBER(4,2) NOT NULL,
    unit_measure VARCHAR2(10) NOT NULL,
    qty_on_hand NUMBER(8) NOT NULL,
    reorder_level NUMBER(8) NOT NULL,
    sell_price NUMBER(8,2) NOT NULL,
    cost_price NUMBER(8,2) NOT NULL,
    CHECK (sell_price > 0),
    CHECK (cost_price > 0)
);
INSERT INTO product_master VALUES('P00001','1.44
Floppies',5,'Piece',100,20,525,500);
INSERT INTO product_master
VALUES('P03453','Monitors',6,'Piece',10,3,12000,11280);
INSERT INTO product_master VALUES('P06734','Mouse',5,'Piece',20,5,1050,1000);
INSERT INTO product_master VALUES('P07865','1.22
Floppies',5,'Piece',100,20,525,500);
INSERT INTO product_master
VALUES('P07868','Keyboard',2,'Piece',10,3,3150,3050);
INSERT INTO product_master VALUES('P07885','CD
Drive',2.5,'Piece',10,3,5250,5100);
INSERT INTO product_master VALUES('P07965','540
HDD',4,'Piece',10,3,8400,8000);
INSERT INTO product_master VALUES('P07975','1.44
Drive',5,'Piece',10,3,1050,900);
INSERT INTO product_master VALUES('P08865','1.22
Drive',5,'Piece',2,3,1025,850);

CREATE TABLE salesman_master (
    salesman_no VARCHAR2(6) PRIMARY KEY CHECK (salesman_no LIKE 'S%'),
    salesman_name VARCHAR2(30) NOT NULL,
    address1 VARCHAR2(30) NOT NULL,
    address2 VARCHAR2(30),
    city VARCHAR2(20),
    pincode NUMBER(8),
    state VARCHAR2(20),
    sal_amt NUMBER(8,2) NOT NULL,
```

```
    CHECK (sal_amt > 0)
);
INSERT INTO salesman_master
VALUES('S001','Kiran','A/14','Worli','Bombay',400002,'Maharastra',3000);
INSERT INTO salesman_master
VALUES('S002','Manish','65','Nariman','Bombay',400001,'Maharastra',3000);
INSERT INTO salesman_master VALUES('S003','Ravi','P-
7','Bandra','Bombay',400032,'Maharastra',3000);
INSERT INTO salesman_master
VALUES('S004','Asish','A/5','Juhu','Bombay',400044,'Maharastra',3000);

CREATE TABLE sales_order (
    order_no VARCHAR2(6) PRIMARY KEY,
    order_date DATE,
    client_no VARCHAR2(6),
    salesman_no VARCHAR2(6),
    delivery_type CHAR(1) DEFAULT 'F',
    bill_y_n CHAR(1),
    delivery_date DATE,
    order_status VARCHAR2(10),

    FOREIGN KEY (client_no) REFERENCES client_master(client_no),
    FOREIGN KEY (salesman_no) REFERENCES salesman_master(salesman_no),

    CHECK (order_no LIKE 'O%'),
    CHECK (delivery_type IN ('P','F')),
    CHECK (delivery_date >= order_date),
    CHECK (order_status IN
('InProcess','Fullfilled','BackOrder','Cancelled'))
);
INSERT INTO sales_order VALUES('O19001',TO_DATE('12-JAN-96','DD-MON-
YY'),'C001','S001','F','N',TO_DATE('20-JAN-96','DD-MON-YY'),'InProcess');
INSERT INTO sales_order VALUES('O19002',TO_DATE('25-JAN-96','DD-MON-
YY'),'C002','S002','P','N',TO_DATE('27-JAN-96','DD-MON-YY'),'BackOrder');
INSERT INTO sales_order VALUES('O46865',TO_DATE('18-FEB-96','DD-MON-
YY'),'C003','S003','F','Y',TO_DATE('20-FEB-96','DD-MON-YY'),'Fullfilled');
INSERT INTO sales_order VALUES('O19003',TO_DATE('03-APR-96','DD-MON-
YY'),'C001','S001','F','Y',TO_DATE('07-APR-96','DD-MON-YY'),'Fullfilled');
INSERT INTO sales_order VALUES('O46866',TO_DATE('20-MAY-96','DD-MON-
YY'),'C004','S002','P','N',TO_DATE('22-MAY-96','DD-MON-YY'),'Cancelled');
INSERT INTO sales_order VALUES('O19008',TO_DATE('24-MAY-96','DD-MON-
YY'),'C005','S004','F','N',TO_DATE('26-MAY-96','DD-MON-YY'),'InProcess');

CREATE TABLE sales_order_details (
    order_no VARCHAR2(6),
    product_no VARCHAR2(6),
    qty_ordered NUMBER(8),
    qty_disp NUMBER(8),
    product_rate NUMBER(10,2),

    FOREIGN KEY (order_no) REFERENCES sales_order(order_no),
    FOREIGN KEY (product_no) REFERENCES product_master(product_no)
);
INSERT INTO sales_order_details VALUES ('O19001','P00001',4,4,525);
INSERT INTO sales_order_details VALUES ('O19001','P07965',2,1,8400);
INSERT INTO sales_order_details VALUES ('O19001','P07885',2,1,5250);
INSERT INTO sales_order_details VALUES ('O19002','P00001',10,0,525);
INSERT INTO sales_order_details VALUES ('O46865','P07868',3,3,3150);
INSERT INTO sales_order_details VALUES ('O46865','P07885',3,1,5250);
INSERT INTO sales_order_details VALUES ('O46865','P00001',10,10,525);
INSERT INTO sales_order_details VALUES ('O46865','P03453',4,4,1050);
INSERT INTO sales_order_details VALUES ('O19003','P03453',2,2,1050);
INSERT INTO sales_order_details VALUES ('O19003','P06734',1,1,12000);
INSERT INTO sales_order_details VALUES ('O46866','P07965',1,0,8400);
```

```
INSERT INTO sales_order_details VALUES ('O46866','P07975',1,0,1050);
INSERT INTO sales_order_details VALUES ('O19008','P00001',10,5,525);
INSERT INTO sales_order_details VALUES ('O19008','P07975',5,3,1050);

COMMIT;
```

DBMS Lab Queries (Questions 1 to 13)

Question 1:

Find the names of all clients having ‘a’ as the second letter in their names. **Query:**

```
SELECT name FROM client_master WHERE name LIKE '_a%';
```

Question 2:

Find out the clients who do not stay in a city whose first letter is ‘B’. **Query:**

```
SELECT * FROM client_master WHERE city NOT LIKE 'B%';
```

Question 3:

List the names and city of all clients who have exactly 12 characters in length and starts with ‘I’. **Query:**

```
SELECT name, city FROM client_master WHERE name LIKE 'I%' AND
LENGTH(name)=12;
```

Question 4:

Find the list of all clients who stay in ‘Bombay’ or ‘Delhi’. **Query:**

```
SELECT * FROM client_master WHERE city IN ('Bombay','Delhi');
```

Question 5:

Print the list of all clients whose bal_due is greater than value 10,000. **Query:**

```
SELECT * FROM client_master WHERE balance_due > 10000;
```

Question 6:

Print the information from sales_order table for orders placed in the month of January. **Query:**

```
SELECT * FROM sales_order WHERE EXTRACT(MONTH FROM order_date)=1;
```

Question 7:

Display the order information for client_no ‘C001’ and ‘C002’. **Query:**

```
SELECT * FROM sales_order WHERE client_no IN ('C001','C002');
```

Question 8:

Find products whose selling price greater than 2000 and less than 5000. **Query:**

```
SELECT * FROM product_master WHERE sell_price > 2000 AND sell_price < 5000;
```

Question 9:

Find products whose selling price is more than 1500. Calculate a new selling price as original selling price*1.15. Rename the new column as New_price. Query:**

```
SELECT product_no, description, sell_price, sell_price*1.15 AS New_price
FROM product_master WHERE sell_price > 1500;
```

Question 10:

List the names, city and state of clients who are not in the state of 'Maharastra'. **Query:**

```
SELECT name, city, state FROM client_master WHERE state <> 'Maharastra';
```

Question 11:

Display the month (in alphabets) and date when the order must be delivered. **Query:**

```
SELECT TO_CHAR(delivery_date,'MONTH') AS Month, TO_CHAR(delivery_date,'DD')
AS Date_No FROM sales_order;
```

Question 12:

Display the Order_date in the format 'DD-Month-YY'. **Query:**

```
SELECT TO_CHAR(order_date,'DD-MONTH-YY') FROM sales_order;
```

Question 13:

Find the date, 15 days after today's date. **Query:**

```
SELECT SYSDATE + 15 FROM dual;
```

DBMS Lab Queries (Questions 14 to 27)

Question 1:

Count the total number of orders. **Query:**

```
SELECT COUNT(*) AS total_orders FROM sales_order;
```

Question 2:

Calculate the average price of all the products. **Query:**

```
SELECT AVG(sell_price) AS avg_price FROM product_master;
```

Question 3:

Count the number of products having price greater than or equal to 1500. **Query:**

```
SELECT COUNT(*) AS total_products FROM product_master WHERE sell_price >=
1500;
```

Question 4:

Determine the maximum and minimum product prices. Rename the output as max_price and min_price respectively. **Query:**

```
SELECT MAX(sell_price) AS max_price, MIN(sell_price) AS min_price FROM
product_master;
```

Question 5:

Change the City of the Client_no ‘C005’ to ‘Madras’. **Query:**

```
UPDATE client_master SET city='Madras' WHERE client_no='C005';
```

Question 6:

Change the Bal_due of Client_no ‘C005’ to Rs.3000/-. **Query:**

```
UPDATE client_master SET balance_due=3000 WHERE client_no='C005';
```

Question 7:

Delete from client_master where the column state holds the value 'Tamilnadu'. **Query:**

```
DELETE FROM client_master WHERE state='TamilNadu';
```

Question 8:

Add a column called 'Telephone' of data type number and size 10 in the table client_master. **Query:**

```
ALTER TABLE client_master ADD telephone NUMBER(10);
```

Question 9:

Change the size of data type Pin_code to 10 in the table client_master. **Query:**

```
ALTER TABLE client_master MODIFY pincode NUMBER(10);
```

Question 10:

Drop the column Address2 from the table client_master. **Query:**

```
ALTER TABLE client_master DROP COLUMN address2;
```

Question 11:

Create another table client_master_duplicate with the same structure of client_master (without copying data). **Query:**

```
CREATE TABLE client_master_duplicate AS SELECT * FROM client_master WHERE
1=2;
```

Question 12:

Insert the data into client_master_duplicate table from client_master table. **Query:**

```
INSERT INTO client_master_duplicate SELECT * FROM client_master;
```

Question 13:

Rename the table client_master_duplicate to c_master. **Query:**

```sql
RENAME client_master_duplicate TO c_master;
```

Question 14:

Destroy the table c_master with its data. **Query:**

```sql
DROP TABLE c_master;
```

Oracle Default Tables (EMP, DEPT) Queries 1 to 14

Question 1:

Display the names of all employees right aligning them to 15 characters. **Query:**

```
SELECT LPAD(ename,15) FROM emp;
```

Question 2:

Display the names of all employees padding them to the right up to 15 characters with '*. *Query:**

```
SELECT RPAD(ename,15,'*') FROM emp;
```

Question 3:

Find the details of all the managers in department 10 and all clerks in department 20 and all employees who are neither managers nor clerks but whose salary is more than or equal to 2000. **Query:**

```
SELECT * FROM emp
WHERE (job='MANAGER' AND deptno=10)
   OR (job='CLERK' AND deptno=20)
   OR (job NOT IN ('MANAGER','CLERK') AND sal>=2000);
```

Question 4:

List all the employees who have joined between 01/02/81 and 31/08/81. **Query:**

```
SELECT * FROM emp
WHERE hiredate BETWEEN TO_DATE('01-02-81','DD-MM-YY') AND
TO_DATE('31-08-81','DD-MM-YY');
```

Question 5:

List all the employees who were joined as manager during 1981. **Query:**

```
SELECT * FROM emp
WHERE job='MANAGER' AND TO_CHAR(hiredate,'YYYY')='1981';
```

Question 6:

List the employees whose salaries are 800, 1600 or 2450. **Query:**

```
SELECT * FROM emp WHERE sal IN (800,1600,2450);
```

Question 7:

List the names of all employees who are either clerks or salesman or analyst. **Query:**

```
SELECT ename FROM emp WHERE job IN ('CLERK','SALESMAN','ANALYST');
```

Question 8:

List the total number of employees and the average salaries of the different departments. **Query:**

```
SELECT deptno, COUNT(*) AS total_emp, AVG(sal) AS avg_sal FROM emp GROUP BY
deptno;
```

Question 9:

Calculate the average salary of all employees whose department is 30. **Query:**

```
SELECT AVG(sal) AS avg_salary FROM emp WHERE deptno=30;
```

Question 10:

Calculate the minimum salary earned by clerks. **Query:**

```
SELECT MIN(sal) AS min_salary FROM emp WHERE job='CLERK';
```

Question 11:

Calculate the maximum salary earned by salesman. **Query:**

```
SELECT MAX(sal) AS max_salary FROM emp WHERE job='SALESMAN';
```

Question 12:

Find the names of those employees whose immediate boss is in different department. **Query:**

```
SELECT e.ename FROM emp e, emp m WHERE e.mgr=m.empno AND e.deptno<>m.deptno;
```

Question 13:

Calculate the number of employees who are not getting any commission. **Query:**

```sql
SELECT COUNT(*) AS total_emp FROM emp WHERE comm IS NULL;
```

Question 14:

Find the department not having any employee. **Query:**

```sql
SELECT * FROM dept WHERE deptno NOT IN (SELECT deptno FROM emp);
```

Oracle EMP / DEPT Advanced Queries 1 to 12

Question 1:

List all the employee names, dept name and the city, in department name order. **Query:**

```
SELECT e.ename, d.dname, d.loc FROM emp e, dept d WHERE e.deptno=d.deptno
ORDER BY d.dname;
```

Question 2:

List all employees working in Dallas in descending order of salary. **Query:**

```
SELECT e.* FROM emp e, dept d WHERE e.deptno=d.deptno AND d.loc='DALLAS'
ORDER BY e.sal DESC;
```

Question 3:

List employee name, department name, job and location of all employees who work in DALLAS. **Query:**

```
SELECT e.ename, d.dname, e.job, d.loc FROM emp e, dept d WHERE
e.deptno=d.deptno AND d.loc='DALLAS';
```

Question 4:

List the employee name, salary, PF, HRA, DA and gross salary; order the result in ascending order of gross. **Query:**

```
SELECT ename, sal,
       sal*0.10 AS PF,
       sal*0.60 AS HRA,
       sal*0.40 AS DA,
       sal + sal*0.10 + sal*0.60 + sal*0.40 AS gross
FROM emp
ORDER BY gross ASC;
```

Question 5:

Display names and salary of all the employees who report to KING. **Query:**

```
SELECT ename, sal FROM emp WHERE mgr=(SELECT empno FROM emp WHERE
ename='KING');
```

Question 6:

List all employees who work in DALLAS and earn more than any employee working in Chicago. **Query:**

```
SELECT e.ename, e.sal FROM emp e, dept d WHERE e.deptno=d.deptno AND
d.loc='DALLAS' AND e.sal > ANY (SELECT e2.sal FROM emp e2, dept d2 WHERE
e2.deptno=d2.deptno AND d2.loc='CHICAGO');
```

Question 7:

List all employees who work in the same post as Smith. **Query:**

```
SELECT * FROM emp WHERE job=(SELECT job FROM emp WHERE ename='SMITH');
```

Question 8:

Find the job with the highest average salary. **Query:**

```
SELECT job FROM (SELECT job, AVG(sal) avg_sal FROM emp GROUP BY job ORDER BY
avg_sal DESC) WHERE ROWNUM=1;
```

Question 9:

List the top 10 earners in the company. **Query:**

```
SELECT * FROM (SELECT * FROM emp ORDER BY sal DESC) WHERE ROWNUM<=10;
```

Question 10:

Display the names of all employees replacing 'A' with 'a'. **Query:**

```
SELECT REPLACE(ename,'A','a') FROM emp;
```

Question 11:

Show the salary of all the employees rounding it to the nearest Rs.1000/-. **Query:**

```
SELECT ename, ROUND(sal,-3) AS rounded_salary FROM emp;
```

Question 12:

Show the first three and last three characters of the names of all the employees. **Query:**

```sql
SELECT ename, SUBSTR(ename,1,3) AS first_three, SUBSTR(ename,-3) AS
last_three FROM emp;
```

DBMS View Experiment (All in One File)

Step 1: Create Table client_master

```
CREATE TABLE client_master (
    client_no VARCHAR2(8) PRIMARY KEY,
    name VARCHAR2(20) NOT NULL,
    address1 VARCHAR2(20) NOT NULL,
    address2 VARCHAR2(20),
    city VARCHAR2(15),
    state VARCHAR2(15),
    pincode VARCHAR2(8),
    bal_due NUMBER(8,3)
);
```

Step 2: Insert Records

```
INSERT INTO client_master VALUES
('C0000001','Ravi','A-10','Park','Delhi','Delhi','110001',1000);
INSERT INTO client_master VALUES
('C0000002','Asha','B-20','Lake','Mumbai','Maharastra','400001',2000);
INSERT INTO client_master VALUES
('C0000003','Neha','C-30','Road','Kolkata','WB','700001',3000);
COMMIT;
```

Step 3: Create View vw_client_master

```
CREATE VIEW vw_client_master AS
SELECT client_no, name, address1, bal_due FROM client_master;
```

Step 4: Insert Through View

```
INSERT INTO vw_client_master VALUES ('C0000004','Kiran','D-40',1500);
INSERT INTO vw_client_master VALUES ('C0000005','Manish','E-50',2500);
INSERT INTO vw_client_master VALUES ('C0000006','Puja','F-60',3500);
COMMIT;
```

Step 5: Update Through View

```
UPDATE vw_client_master SET bal_due=5000 WHERE client_no='C0000004';
COMMIT;
```

Step 6: Delete Through View

```
DELETE FROM vw_client_master WHERE client_no='C0000006';
COMMIT;
```

Step 7: Check Base Table

```
SELECT * FROM client_master;
```

Step 8: Create Table sales_order

```
CREATE TABLE sales_order (
    order_no VARCHAR2(6) PRIMARY KEY,
    order_date DATE,
    client_no VARCHAR2(8),
    order_status VARCHAR2(15)
);
```

Step 9: Create Table sales_order_details

```
CREATE TABLE sales_order_details (
    order_no VARCHAR2(6),
    product_no VARCHAR2(6),
    qty_ordered NUMBER(8)
);
```

Step 10: Insert Sample Data

```
INSERT INTO sales_order VALUES ('O10001',SYSDATE,'C0000001','Backorder');
INSERT INTO sales_order VALUES ('O10002',SYSDATE,'C0000002','Fullfilled');
INSERT INTO sales_order_details VALUES ('O10001','P001',5);
INSERT INTO sales_order_details VALUES ('O10002','P002',3);
COMMIT;
```

Step 11: Create View vw_sales_det

```
CREATE VIEW vw_sales_det AS
SELECT s.client_no, s.order_no, s.order_date,
       d.product_no, d.qty_ordered, s.order_status
FROM sales_order s, sales_order_details d
WHERE s.order_no=d.order_no
AND s.order_status='Backorder';
```

Step 12: Display View

```
SELECT * FROM vw_sales_det;
```

Step 13: Update Base Table (Safe Method)

```
UPDATE sales_order SET client_no='C0000003' WHERE order_no='O10001';
COMMIT;
```

Step 14: Delete Base Table Record

```
DELETE FROM sales_order_details WHERE order_no='O10001';
COMMIT;
```

Step 15: Drop Views

```
DROP VIEW vw_client_master;
DROP VIEW vw_sales_det;
```